Foundations of Attention mechanism neural network

[Attention mechanism neural network]

$$a^{(l)} = \sigma! \left(W^{(l)} a^{(l-1)} + b^{(l)}\right)$$

1.0 Content Block

Positional encoding Since the Transformer does not rely on recurrence or convolution of the text in order to perform encoding and decoding, the paper relied on the use of sine and cosine wave functions to encode the position of the token into the embedding. The methods introduced in the paper are discussed below:

2.0 Content Block

P E (p o s , 2 i + 1) = cos ■ (p o s / 10000 2 i / d m o d e l) {\displaystyle PE_{({\rm {pos}},2i+1)}=\cos({\rm {pos}}/{10000}^{2i/d_{\rm {model}}})}

3.0 Content Block

where Q {\displaystyle Q} , K {\displaystyle K} , V {\displaystyle V} are respectively the query, key, value matrices, and d k {\displaystyle d_{k}} is the dimension of the values. Since the model relies on Query (Q), Key (K), and Value (V) matrices that come from the same source (i.e., the input sequence or context window), this eliminates the need for RNNs, completely ensuring parallelizability for the architecture. This differs from the original form of the Attention mechanism introduced in 2014. Additionally, the paper also discusses the use of an additional scaling factor that was found to be most effective with respect to the dimension of the key vectors (represented as d k {\displaystyle d_{k}} and initially set to 64 within the paper) in the manner shown above. In the specific context of translation, which the paper focused on, the Query and Key matrices are usually represented in embeddings corresponding to the source language, while the Value matrix corresponds to the target language.

4.0 Content Block

Dataset - The English-to-German translation model was trained on the 2014 WMT (Workshop on Statistical Machine Translation) English-German dataset, consisting of nearly 4.5 million sentences derived from TED Talks and high-quality news articles. A separate translation model was trained on the much larger 2014 WMT English-French dataset, consisting of 36 million sentences. Both datasets were encoded with byte-pair encoding. Hardware - The models were trained using 8 NVIDIA P100 GPUs. The base models were trained for 100,000 steps, and the big models were trained for 300,000 steps - each step taking about 0.4 seconds to complete for the base models and 1.0 seconds for the big models. The base model was trained for a total of 12 hours, and the big model was trained for a total of 3.5 days. Both the base and big models outperform the 2017 state-of-the-art in both English-German and English-French, while achieving the comparatively lowest training cost. Hyperparameters and regularization - For their 100M-parameter Transformer model, the authors increased the learning rate linearly for the first 4000 (warmup) steps and decreased it proportionally to the inverse square root of the current step number. Dropout layers were applied to the output of each sub-layer before normalization, the sums of the embeddings, and the positional encodings. The dropout rate was set to 0.1. Label smoothing was applied with a value of 0.1, which "improves accuracy and BLEU score".

5.0 Content Block

"Attention Is All You Need" is a 2017 research paper in machine learning authored by eight scientists and engineers working at Google. The paper introduced a new deep learning architecture known as the transformer, based on the attention mechanism proposed in 2014 by Bahdanau et al. The transformer approach it describes has become the main architecture of a wide variety of artificial intelligence, including large language models. At the time, the focus of the research was on improving Seq2seq techniques for machine translation, but the authors go further in the paper, foreseeing the technique's potential for other tasks like question answering and what is now known as multimodal generative AI. Some early examples that the team tried their Transformer architecture on included English-to-German translation, generating Wikipedia articles on "The Transformer", and parsing. These convinced the team that the Transformer is a general-purpose language model, and not just good for translation. As of 2025, the paper has been cited more than 173,000 times, placing it among the top ten most-cited papers of the 21st century. After the paper was published by Google, each of the authors left the company to join other companies or to found startups.

6.0 Content Block

Predecessors For many years, sequence modelling and generation was done by using plain recurrent neural networks (RNNs). A well-cited early example was the Elman network (1990). In theory, the information from one token can propagate arbitrarily far down the sequence, but in practice the vanishing-gradient problem leaves the model's state at the end of a long sentence without precise, extractable information about preceding tokens. A key breakthrough was LSTM (originally described in a 1995 technical report and formally published in 1997), an RNN that introduced gating mechanisms to mitigate the vanishing gradient problem, allowing efficient learning of long-sequence modelling. One key architectural element was the use of multiplicative gating units, in which the outputs of some neurons modulate the outputs of others. These multiplicative units are conceptually distinct from the additive attention mechanism later introduced for sequence-to-sequence models. Neural networks using multiplicative units were later called sigma-pi networks or higher-order networks. LSTM became the standard architecture for long sequence modelling until the 2017 publication of transformers. However, LSTM still used sequential processing, like most other RNNs. Specifically, RNNs operate one token at a time from first to last; they cannot operate in parallel over all tokens in a sequence. Modern transformers overcome this problem, but unlike RNNs, they require computation time that is quadratic in the size of the context window. The linearly scaling fast weight controller (1992) learns to compute a weight matrix for further processing depending on the input. One of its two networks has "fast weights" or "dynamic links" (1981). A slow neural network learns by gradient descent to generate keys and values for computing the weight changes of the fast neural network which computes answers to queries. This was later shown to be equivalent to the unnormalized linear transformer.

7.0 Content Block

While the primary focus of the paper at the time was to improve machine translation, the paper also discussed the use of the architecture on English Constituency Parsing, both with limited and large-sized datasets, achieving a high-score without specific tuning for the task indicating the promising nature of the model for use in a wide-variety of general purpose of seq2seq tasks.